You-Ting Wu (he/him)

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Research Interests/Keywords

dynamical systems and climate tipping points, uncertainty quantification in ensemble projections, attribution of forced climate responses, Arctic climate change, energetic feedbacks, climate modeling, machine learning in climate change

Education

Sept 2024–present | **Columbia University**, New York, NY, USA
Ph.D., *Applied Mathematics* (expected 2028)
— Advisors: Lorenzo Polvani and Michael Previdi

Sept 2018–July 2023 | **National Taiwan University**, Taipei, Taiwan
Double BSc., *Mechanical Engineering* and *Atmospheric Sciences*

Awards

Jun 2026 | Government Scholarship to Study Abroad, *Ministry of Education, Taiwan*

Jan 2026 | Nominated for Membership, *Sigma Xi (The Scientific Research Honor Society), USA*

June 2023 | Excellent Undergraduate Research Project Award, *Ministry of Science and Technology, Taiwan*

June 2022 | Undergraduate Research Project Grants, *Ministry of Science and Technology, Taiwan*

Publications

5. Previdi, M., **Wu, Y.-T.**, Polvani, L. M., England, M. R., Sigmond, M., and Liang, Y.-C. Substantial twentieth-century global sea level rise caused by ozone-depleting substances. Submitted to *Journal of Climate* (under review).

4. **Wu, Y.-T.**, Previdi, M., Romanou, A., Lerner, P., and Polvani, L.M. (2026). Arctic Climate Tipping Points under Delayed CO2 Emissions Mitigation. *Communications Earth & Environment* (in press).

3. Liang, Y.-C., Polvani, L. M., Previdi, M., Dong, Y., **Wu, Y.-T.**, England, M.R., and Griffith, S. M. (2026). The importance of non-CO2 greenhouse gases to Arctic warming and sea-ice loss. *Geophysical Research Letters*, 53, e2026GL122867.

2. **Wu, Y.-T.**, Liang, Y.-C., Previdi, M., Polvani, L. M., England, M. R., Sigmond, M., and Lo, M.-H. (2024). Stronger Arctic Amplification from Anthropogenic Aerosols than from Greenhouse Gases. *npj Climate and Atmospheric Science* 7 (1), 142.

1. **Wu, Y.-T.**, Liang, Y.-C., Kuo Y.-N., Lehner, F., Polvani, L. M., Previdi, M., Lo, M.-H., Lan, C.-W. (2023). Exploiting SMILEs and the CMIP5 archive to understand Arctic climate change seasonality and uncertainty. *Geophysical Research Letters*, 50(2), e2022GL100745.

Research Experiences

Sept 2024–present | **Graduate Research Assistant**, Columbia University
New York, NY, USA
- Modeled nonlinear, bifurcating dynamics of the Arctic climate system under CO_2 forcing using large-ensemble Earth system model simulations; identified a critical tipping threshold via regime-transition analysis
- Compiled, ran, and modified the Community Earth System Model (CESM2), a Fortran codebase parallelized with MPI/OpenMP, on high-performance computing clusters to generate large-ensemble simulations
- Built CNN and LSTM neural-network emulators to predict spatiotemporal cold-spell extremes from Earth system model output; evaluated out-of-sample generalization across unseen forcing scenarios
- Quantified sea-level rise and Arctic sea-ice loss under different anthropogenic forcings (CO_2 vs. aerosols) using coupled climate and numerical models

June 2022–Aug 2022 | **Visiting Student Intern**, Lamont-Doherty Earth Observatory, Columbia University
Palisades, NY, USA
- Applied bootstrap resampling and statistical significance testing to large-ensemble climate simulations to quantify the robustness of an aerosol-driven Arctic amplification signal against internal variability

June 2021–July 2024 | **Undergraduate Researcher**, Polar Climate Research Group, National Taiwan University
Taipei, Taiwan
- Implemented physics-based climate models to identify the impacts of different anthropogenic forcings on climate change
- Processed geospatial data to quantify key climate indices; translated complex data into accessible visuals to understand projected Arctic climate

Invited Talks and Seminars

June 2025 | (invited) Polar Teatime Seminar, National Central University, Taoyuan, Taiwan
Stronger Arctic amplification from anthropogenic aerosols than from greenhouse gases

Oct 2022 | Seminar at Earth and Life Science Department, University of Taipei, Taipei, Taiwan (with Yu-Chiao Liang)
The Arctic Amplification, and its Seasonal Migration, under Greenhouse Gases Forcings

Apr 2022 | Seminar at Institute of Hydrological and Oceanic Sciences, National Central University, Taoyuan, Taiwan (with Yu-Chiao Liang)
The Arctic Amplification, and its Seasonal Migration, under Greenhouse Gases Forcings

Mar 2022 | Seminar at Institute of Oceanography, NTU, Taipei, Taiwan (with Yu-Chiao Liang)
The Arctic Amplification, and its Seasonal Migration, under Greenhouse Gases Forcings

Conference Presentations

Dec 2025 | (poster) AGU25 Annual Meeting, New Orleans, LA
Arctic Climate Tipping Points under Delayed CO2 Emissions Mitigation

May 2025 | (oral) AMS 2025 Denver Summit, Denver, CO
Stronger Arctic amplification from anthropogenic aerosols than from greenhouse gases

Dec 2024 | (poster) AGU24 Annual Meeting, Washington, D.C.
Stronger Arctic amplification from anthropogenic aerosols than from greenhouse gases

Aug 2022 | (oral) AMS 2022 Collective Madison Meeting, Madison, WI
Decomposing Projection Uncertainty for the Seasonal Evolution of Arctic Amplification

Aug 2022 | (oral) The 19th Annual Meeting of Asia Oceania Geosciences Society, virtual
Decomposing Projection Uncertainty for the Seasonal Evolution of Arctic Amplification

June 2022 | (oral) The 27th Annual CESM Workshop, virtual
Decomposing Projection Uncertainty for the Seasonal Evolution of Arctic Amplification

Sept 2021 | (poster) Undergraduate Summer Research Program, National Taiwan University, Taipei, Taiwan
Seasonal Evolution and Uncertainty of Arctic Amplification in Multi-Model Large Ensembles

Aug 2021 | (poster) 2021 Climate Hotpot in Action Forum, virtual
Seasonal Evolution and Uncertainty of Arctic Amplification in Multi-Model Large Ensembles

Teaching Experiences

Teaching Assistant, Columbia University (6×)

Fall 2026 | Research Computing in Earth Science; Scientific Programming for Sustainability Science

Fall 2025 | Introduction to Atmospheric Science

Spring 2025 | Partial Differential Equations; Statistics, Data Analysis, and Coding for Sustainability Science

Fall 2024 | Physics of Fluids

Teaching Assistant, National Taiwan University (6×)

Spring 2023 | Numerical Analysis; Workshop for Earth System Model; Atmospheric Radiation

Fall 2022 | Applied Mathematics I

Spring 2022 | Numerical Analysis; Middle-high Latitude Stratosphere-Troposphere Dynamics

Professional Service

2026 | Peer Reviewer, *Nature Communications*

Working Experiences

July 2023–Sept 2023 | **Software Engineering Intern**, Formosa Climate Smart Service Limited
Taipei, Taiwan

- Built database infrastructure to convert engineering inputs into quantifiable Energy Use Intensity (EUI) metrics
- Developed front-end platform to visualize building EUI reports
- Designed a workflow to integrate geospatial data and engineering drawings into 3D building-EUI interfaces using AutoCAD, ArcGIS, and Python

Skills

Applied Mathematics	Numerical analysis, linear algebra, nonlinear dynamical systems, time series analysis, bootstrap resampling and uncertainty quantification
Machine Learning	Deep learning (CNN, LSTM) for spatiotemporal emulation of climate model output; out-of-sample evaluation across unseen forcing scenarios
Programming & HPC	Python (SciPy, PyTorch, scikit-learn), MATLAB, Fortran (MPI, OpenMP), C/C++, GrADS, HTML, JavaScript; high-performance computing on NCAR Derecho and Casper
Computer-aided Design	Autodesk AutoCAD, Autodesk Inventor
Computational Fluid Analysis	ANSYS Fluent
Language	Mandarin (native), English (fluent), German (basic)